Project Configuration Report

NT-3-Light

Target Device:

Target Part: EFR32MG21A020F1024IM32
Boards:

Target SDK

Simplicity SDK Suite v2025.6.0: Bluetooth 10.0.0, Bluetooth Mesh 9.0.0, Connect 4.1.0.0, EmberZNet 8.2.0.0, Micrium OS Kernel 5.18.00, OpenThread 2.7.0.0 (GitHub-fb0446f53), Platform 5.2.0.0, RAIL 2.19.0, Sidewalk 2.5.0, Silicon Labs Matter 2.6.0-1.4, USB 1.5.0.0, Wi-Fi SDK 3.5.0, Wi-SUN 2.6.0.0, Z-Wave SDK 7.24.0.0
SDK Version: 2025.6.0

Extensions

• Sidewalk 2.5.0
• WiSeConnect 3 3.5.0
• Silicon Labs Matter 2.6.0

Project Generators

• Simplicity IDE Project

Installed Software Components:

• Platform

- Device

• EFR32MG21

EFR32MG21A020F1024IM32
Sources: startup_efr32mg21.c, system_efr32mg21.c
Configurable: true
Configs: device.yaml

- Driver

• I2C

I2CSPM
Configurable: true
Configs: sl_i2cspm_I2C_config.h

• PWM

PWM
Configurable: true
Configs: sl_pwm_init_pwm_config.h

- *Peripheral*
  USART
  Sources: sl_hal_usart.c
  Configurable: false

  • HAL
    TIMER
    Sources: sl_hal_timer.c
    Configurable: false

    WDOG
    Sources: sl_hal_wdog.c
    Configurable: false

- *Toolchain*
  Enable Link time optimization
  Configurable: false

• Services

- *Clock Manager*
  Clock Manager
  Sources: sl_clock_manager_init.c
  Configurable: true
  Configs: sl_clock_manager_oscillator_config.h, sl_clock_manager_ext_flash_config.h, sl_clock_manager_tree_config.h

- *Clocks*
  HFXO Manager
  Sources: sl_hfxo_manager_hal_s2.c, sli_hfxo_manager_internal.h, sl_hfxo_manager.c
  Configurable: true
  Configs: sl_hfxo_manager_config.h

- *Command Line Interface*
  CLI Instance(s)
  Configurable: true
  Configs: sl_cli_config_example.h

- *IO Stream*
  IO Stream: Recommended Stream for a console
  Configurable: false

  • Driver
    IO Stream: USART
    Configurable: true
    Configs: sl_iostream_usart_UART_config.h

- *System Setup*

  System Setup (sl_main)
  Configurable: true
  Configs: sl_main_start_task_config.h

• Zigbee

- *Bootloader*

  Zigbee Application Bootloader
  Sources: app_properties.c
  Configurable: false

- *Cluster Library*

  ZCL Framework Core
  Sources: process-cluster-message.c, af-node.c, library.c, af-soc-common.c, af-common.c, af-event.c, zcl-util.c, util.c, global-callback.c, option-cli.c, process-global-message.c, security-cli.c, zcl-framework-core-cb.c, af-trust-center.c, attribute-size.c, attribute-storage.c, zigbee-device-library.c, global-other-callback.c, message.c, zdo-cli.c, service-discovery-soc.c, time-util.c, sl_zigbee_command_interpreter.c, print.c, service-discovery-common.c, zigbee-device-common.c, attribute-table.c, print-formatter.c, zcl-cli.c, crypto-state.c, network-cli.c, af-soc.c, multi-network.c, af-security-common.c, core-cli.c, client-api.c
  Configurable: true
  Configs: zcl-framework-core-config.h

  • Common

    Basic Server Cluster
    Sources: basic-cb.c, basic.c
    Configurable: false

    Groups Client Cluster
    Sources: groups-client.c
    Configurable: false

    Identify Cluster
    Sources: identify-cb.c, identify-cli.c, identify.c
    Configurable: false

    Level Control Server Cluster
    Sources: level-control-cb.c, level-control.c
    Configurable: true
    Configs: level-control-config.h

    Reporting
    Sources: reporting-cb.c, reporting.c, reporting-cli.c, reporting-default-configuration.c
    Configurable: true
    Configs: reporting-config.h

    Scenes Client Cluster
    Sources: scenes-client.c

Configurable: false

• Home Automation

Color Control Cluster Server
Sources: color-control-server.c, color-control-server-cb.c
Configurable: true
Configs: color-control-server-config.h

On/Off Server Cluster
Sources: on-off.c, on-off-cb.c
Configurable: false

- Green Power

Green Power Client
Sources: green-power-client.c
Configurable: true
Configs: green-power-client-config.h

Green Power Client CLI
Sources: green-power-client-cli.c, green-power-client.c, green-power-cli.c
Configurable: false

Green Power Common
Sources: gp-util.c, green-power-common.c
Configurable: false

- Network Form and Join

Scan Dispatch
Sources: scan-dispatch.c
Configurable: true
Configs: scan-dispatch-config.h

- Over-The-Air Bootloading

OTA Bootload Cluster Client
Sources: ota-client-signature-verify.c, ota-client-page-request.c, ota-client.c, ota-client-cli.c, ota-client-cb.c
Configurable: true
Configs: ota-client-config.h

OTA Simple Storage EEPROM Driver
Sources: ota-storage-simple-eeprom-cb.c, ota-storage-eeprom-page-erase.c, ota-storage-eeprom-read-modify-write.c, ota-storage-eeprom.c
Configurable: true
Configs: ota-storage-simple-eeprom-config.h

OTA Simple Storage Module
Sources: ota-storage-simple.c, ota-storage-simple-simulation.c
Configurable: false

- *Stack*

Debug Basic
Configurable: false

Green Power
Sources: gp-proxy-table-baremetal-wrapper.c, gp-proxy-table-ipc-command-messages.c, gp-sink-table-baremetal-wrapper.c, gp-types-baremetal-wrapper.c, zigbee_ipc_callback_event_stub.c, gp-sink-table-ipc-command-messages.c, command-handlers-green-power-generated.c, gp-types-ipc-command-messages.c, zigbee_ipc_callback_events.c
Configurable: true
Configs: sl_zigbee_green_power_config.h

Zigbee light link
Sources: zll-api-ipc-callback-events.c, zll-api-baremetal-wrapper.c, zigbee_ipc_callback_event_stub.c, zll-api-baremetal-callbacks.c, zll-api-ipc-command-messages.c, zigbee_ipc_callback_events.c, command-handlers-zll-generated.c
Configurable: true
Configs: sl_zigbee_light_link_config.h

• Pro Core

Pro Stack (Common)
Configurable: true
Configs: sl_zigbee_pro_stack_config.h

- *Utility*

Counters
Sources: counters-cb.c, af-counters.c, counters-ota.c, counters-cli.c, counters-ota-host.c
Configurable: true
Configs: counters-config.h

Debug Print
Sources: sl_zigbee_debug_print.c
Configurable: true
Configs: sl_zigbee_debug_print_config.h

Interpan
Sources: interpan-host.c, interpan.c, interpan-soc.c, interpan-cb.c, interpan-cli.c
Configurable: true
Configs: interpan-config.h

Zigbee Core CLI
Sources: core-cli.c
Configurable: true
Configs: zigbee_core_cli_config.h

Zigbee ZCL CLI
Sources: zigbee-zcl-custom-cluster-cli.c, zigbee-zcl-cli.c, zigbee-zcl-global-cli.c
Configurable: false

- *Zigbee 3.0*

Find and Bind Target
Sources: find-and-bind-target.c, find-and-bind-target-cli.c
Configurable: true
Configs: find-and-bind-target-config.h

Network Steering
Sources: network-steering-cb.c, network-steering.c, network-steering-v2.c, network-steering-cli.c
Configurable: true
Configs: network-steering-config.h

Update TC Link Key
Sources: update-tc-link-key.c, update-tc-link-key-cli.c, update-tc-link-key-cb.c
Configurable: true
Configs: update-tc-link-key-config.h

- *Zigbee Light Link*

ZLL Commissioning Client
Sources: zll-commissioning-client.c, zll-commissioning-client-cb.c
Configurable: true
Configs: zll-commissioning-client-config.h

ZLL Commissioning Common
Sources: zll-commissioning-common.c, zll-commissioning-common-cb.c, zll-commissioning-cli.c
Configurable: true
Configs: zll-commissioning-common-config.h

ZLL Commissioning Server
Sources: zll-commissioning-server-cb.c, zll-commissioning-server.c
Configurable: true
Configs: zll-commissioning-server-config.h

ZLL Utility Client Cluster
Sources: zll-utility-client.c
Configurable: false

Pins:

Peripherals:

Peripheral	Software Component	Custom Peripheral Name
ACMP0		
ACMP1		
CMU	Clock Manager	
GPIO	SWO Debug	
I2C0	I2CSPM (I2C)	
I2C1	I2CSPM (I2C)	
IADC0		
LETIMER0		
LFXO		
MODEM		
PRS Channel ASYNCH0		
PRS Channel ASYNCH1		
PRS Channel ASYNCH10		
PRS Channel ASYNCH11		
PRS Channel ASYNCH2		
PRS Channel ASYNCH3		
PRS Channel ASYNCH4		
PRS Channel ASYNCH5		
PRS Channel ASYNCH6		
PRS Channel ASYNCH7		
PRS Channel ASYNCH8		
PRS Channel ASYNCH9		
PRS Channel SYNCH0		
PRS Channel SYNCH1		
PRS Channel SYNCH2		
PRS Channel SYNCH3		
PTI	RAIL Utility, PTI	
TIMER0	PWM (pwm)	
TIMER1	PWM (pwm)	
TIMER2	PWM (pwm)	
TIMER3	PWM (pwm)	
USART0	IO Stream: USART (UART)	
USART1	IO Stream: USART (UART)	
USART2	IO Stream: USART (UART)	